The optimizer uses a *cost-based algorithm* to compare costs and select the lowest-cost plan. Plan selection is a very interesting and deep topic that you'll also get some experience with.

Besides writing SQL, you'll run statements and commands for PostgreSQL, sometimes using built-in *functions*. Functions enhance your database; there are built-in functions and you may create your own as *User Defined Functions*. Functions can be written using SQL or other programming languages.

Statements you write in PostgreSQL can be classified in a couple of ways. A statement that changes the structure of the database, like creating a table, is called a Data Definition Language (DDL) statement. An example would be `CREATE DATABASE` which is not a full statement and won't run as-is, but meant to be an example.

Another type of statement modifies data and not structure. This statement type is called a Data Manipulation Language (DML) statement. An example is an `INSERT` statement that inserts data into a table. Database users might use these acronyms DDL and DML in conversation to categorize statement types when the categorization is relevant.

PostgreSQL is a transactional database, which means most operations take place inside of a *Transaction*. Transactions are used to group together multiple operations that should occur as one unit. Transactions have an isolated view of data and the level of isolation can be configured, which you'll learn about.

Let's recap some of the terms you've seen so far.

Glossary of Databases Terms



- Relation or `relname` — a table
- Data Manipulation Language (DML) — e.g. `INSERT`, `UPDATE`, and `DELETE` operations
- Data Definition Language (DDL) — e.g. Structure modifications like `ALTER TABLE` or `CREATE INDEX` statements
- Transaction — A unit of work

Ruby on Rails Terminology

Active Record is the Object Relational Mapper (ORM) for Ruby on Rails. Active Record classes are Ruby classes. A class also serves as an interface to a table where model instances are persisted as table rows. Thus an Active Record Ruby class also acts as a persistence interface for application code to write